

Heat transport in chaotic magnetic fields

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1 Create a map

We want to create a map of the magnetic field lines using PyOculus. The magnetic configuration considered is described using slab geometry through

$$\psi = \psi_0(x) + \psi_1(x, y, z) , \quad (1)$$

where ψ_0 describes the equilibrium configuration and ψ_1 is a perturbation of amplitude ϵ with poloidal and toroidal mode numbers m and n :

$$\psi_0(x) = \frac{\iota'}{2}(x - x_0)^2 , \quad (2)$$

$$\psi_1(x, y, z) = \epsilon(x) \cos \alpha \equiv \epsilon(x) \cos(my - nz) . \quad (3)$$

For a system with $x \in [0, 1]$ containing an island at $x_s = 0.5$, as a simple model for a perturbation localized around the rational surface, we choose

$$\epsilon(x) = \delta \cdot x(1 - x) . \quad (4)$$